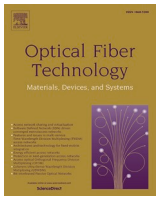
Optical Fiber Technology 81 (2023) 103571



Contents lists available at ScienceDirect

Optical Fiber Technology

journal homepage: www.elsevier.com/locate/yofte



Regular Articles



Link duration-aware shared backup path protection (LD-SBPP) against SRLG failure in LEO-satellite-terrestrial integrated optical networks

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ARTICLE INFO

Keywords:
Low earth orbit (LEO)
Shared risk link group (SRLG)
Survivability
Terrestrial optical network (TON)
Shared backup path protection (SBPP)

ABSTRACT

Large-scale disasters are very destructive to terrestrial optical networks (TONs), and most existing survivability technologies that rely solely on TONs themselves can hardly combat disasters. Satellite networks, as a promising complement to terrestrial systems, can help to improve the survivability of TONs against disasters. However, the time-varying nature of satellite networks poses new challenges to traditional survivability technologies. This paper considers the low earth orbit (LEO) satellite network-assisted shared backup path protection. Since the link-switching consumes a certain amount of time and would cause the delay jitter and the quality of transmission (QoT) fluctuation of the space lightpath, a link duration-aware shared backup path protection (LD-SBPP) scheme is proposed to search space backup lightpaths with the minimum number of link-switching. A pre-calculated snapshot continuous routing and wavelength assignment (P-SCRWA) algorithm is proposed to pre-calculate the routing and reserve spectrum resources at each continuous snapshot. Simulations show that user requests cannot be effectively protected by relying on the TON solely, while the protection can be achieved with the help of LEO satellites. The LD-SBPP scheme has high wavelength utilization efficiency, resulting in lower blocking probability. Besides, the link duration-aware strategy can greatly reduce the number of link-switching along space backup lightpaths.

1. Introduction

With the extensive construction of optical networks, the explosive growth of information carried on optical networks, and the continuous emergence of Internet applications, the hazards of terrestrial optical network (TON) failures have become increasingly severe. In fact, TONs are extremely prone to failures, especially when disasters occur. Natural disasters such as earthquakes, floods, and typhoons, are likely to cause the bending of optical fibers and damage to devices, resulting in network failures. In addition, man-made factors such as regional power outages and electromagnetic pulse attacks may also cause network failures within a certain range. Therefore, survivability is an important aspect of the TON to combat failures. In the TON, many survivability technologies have been studied, such as backup path protection [1], p-cycle protection [2], multipath protection [3–5], and pre-configured polyhedron protection [6]. However, when a disaster occurs, considering the extreme case where a network segment is completely isolated from the TON, these survivability technologies that rely on the network connectivity of the TON become ineffective. Since the survivability technologies relying solely on the TON have encountered challenges, one possible solution is to use the satellite optical network (SON) to protect the TON. Recently, the sixth-generation mobile communication (6G) has obtained comprehensive attention and is widely studied. The 6G supports the seamless communication coverage of the whole time and the entire region with a large number of access users. Over the past decades, tens of thousands of satellites have been launched into space to construct powerful satellite constellations such as Iridium, BeiDou, Starlink, Telesat, HughesNet, etc. The optical wireless communication (OWC) technology has been applied to inter-satellite links and satellite-to-ground links to realize high-rate and large-capacity communication between satellites and between satellites and ground stations (GSs) respectively. In [7], it is reported that the first free space optical communication in Europe has completed several 10 Mbps links between SOTA and MeO optical ground stations at 1549 and 976 nm. In [8], Hiroaki *et al.* reported experimental results of downlinking data from an optical communication terminal attached to the International Space Station to an optical ground station in Greece. Moreover, multiple wavelengths transmission in optical inter-satellite links and optical

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https://doi.org/10.1016/j.yofte.2023.103571
Received 10 August 2023; Received in revised form 27 September 2023; Accepted 22 October 2023
Available online 26 October 2023
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satellite-to-ground links is also researched and verified. In [9], Juraj *et al.* reported on a successful proof-of-concept demonstration of 1.72 Tbit/s transmission using dense wavelength division multiplexing (DWDM) through a turbulent atmosphere in a scenario relevant to a geostationary earth orbit (GEO) uplink. This throughput was achieved by modulating 40 DWDM channels with an uncoded 43.01824 Gbit/s rate per channel. In [10], a high-capacity DWDM technique based low power inter-satellite optical wireless communication (LP-IsOWC) System was proposed for inter-satellite communication. In simulations, 20 DWDM channels each transmitting optical data at 5 Gbps were verified. In references [11–13], the routing and wavelength assignment problem was addressed where the optical inter-satellite link was assumed to have multiple available wavelengths. It can be seen that the SON will play an important role in the future 6G era. Owing to the rapid development of the SON, it can be used as a complement to the TON because it allows global coverage and is not affected by most terrestrial disasters. Several researchers have proposed some failure-resistant technologies using the SON. In fact, the low earth orbit (LEO) satellite system is the focus of current research because of its low latency. The GEO satellite altitude is 35,786 km, assuming a minimum elevation angle of 10 degrees for terrestrial equipment. The maximum distance between terrestrial equipment and satellite is 40,581 km, resulting in a propagation delay of 270.73 ms. While the LEO satellite, with an altitude of 1200 km, has a distance to terrestrial equipment of 3131 km, under the same assumptions, and the propagation delay is 20.89 ms. The LEO scenario is superior to the GEO in this respect. Therefore, this paper studies the implementation of the proposed protection scheme using LEO satellites.

LEO satellites are highly dynamic. LEO satellites run in circular orbits with an altitude of fewer than 2000 km, and their orbital periods are about 100 min. Taking the Iridium system as an example, whose altitude is 780 km and inclination is 86.4 degrees [14], the speed of an Iridium satellite is about 7.5 km per second, and the visibility time of a satellite for a GS is about 10 min assuming a minimum elevation angle of 8.2 degrees. In addition, visibility changes and Doppler drift caused by relative motion between satellites can impact the availability of inter-satellite links (ISLs). Compared to GEO satellites, LEO satellites of high dynamics bring two problems. One of them is the control plane needs to perform a handover when the satellite topology changes, and reconstruct the path and allocate resources for requests. In [15], a resource allocation algorithm based on wavelength division multiplexing (WDM) was proposed. In that paper, link handover that functions to reallocate resources for disconnected requests is executed at the end of each snapshot. There is no guarantee of adequate resources after the handover. This means that requests may be disconnected due to handover. This disconnection is contrary to the objective of survivability. The solution to this problem is pre-calculated, i.e., pre-calculated the routing and reserve spectrum resources at each continuous snapshot. The other problem is that the GEO link duration can be seen as permanent, while the LEO link duration is limited, so the duration of the link needs to be taken into consideration. Choosing a path with a longer duration is likely to reduce the number of link-switching. The handover between two satellites or between a satellite and a GS must conduct the process of acquisition, tracking, and pointing (ATP). Because optical terminals cannot switch as quickly as microwave ISLs, typically requiring nearly 60 s, the ATP delay should not be disregarded [16]. The ATP delay will trigger the re-routing. During the process of the handover, the delay may lead to data loss. Moreover, the ATP process is not always able to be successfully executed. In other words, the handover may fail. Therefore, the space laser link with a long duration should be used as much as possible to reduce the number of link-switching.

In this paper, we study the shared backup path protection for the TON with the assistance of LEO satellites considering disaster conditions. A link duration-aware shared backup path protection (LD-SBPP) scheme is proposed. A pre-calculated snapshot continuous routing and wavelength assignment (P-SCRWA) algorithm is proposed to realize this scheme. The algorithm takes advantage of the predictability of the satellite constellation to calculate the routing and reserve the resources before and after handover in advance. The algorithm also considers the duration of each space laser link to reduce the number of link-switching along space lightpaths. The rest of the paper is organized as follows. Section 2 elaborates on related research progress regarding the routing problem in the SON and existing satellite-assisted survivability technologies. The contributions of this paper are also summarized. Section 3 presents the integrated architecture of the TON and the SON, the disaster modeling, the backup lightpaths in the SON, and the proposed LD-SBPP scheme. The P-SCRWA algorithm is developed in Section 4. The performance of the LD-SBPP scheme is presented and analyzed in Section 5. Finally, Section 6 concludes this paper.

2. Related works

This section first summarizes the latest research progress on the routing problem in the SON. Then, existing satellite assisted survivability technologies are elaborated. At last, the contributions of this paper are summarized.

2.1. The routing problem in the SON

The optical wireless communication technology is applied to inter-satellite and satellite-to-ground links to realize high-rate and large-capacity communication. The SON is being verified and constructed by theoretical research and experiments. In [17], the authors showed that inter-satellite links substantially reduced the temporal variations in latency, added greater resilience to weather, and could yield more than 3x of the throughput achieved without such links. In [18], a framework for simulating and visualizing the network behavior of LEO satellite constellations was presented. In [19], the authors examined, qualitatively and quantitatively, the opportunities and challenges of in-orbit computing. Adding computing hardware to a satellite does not seem prohibitive in terms of weight, volume, and space hardening, but the required power draw could be substantial. In [20], the author built a simulator based on public details from the US Federal Communications Committee (FCC) filings. They evaluated how to use the laser links to provide a network and looked at the routing problem on this network. They concluded that a network built in this manner can provide lower latency communications than any possible terrestrial optical fiber network for communications over distances greater than about 3000 km. In [21], the author investigated using ground-based relays to substitute inter-satellite links to provide low-latency wide-area networking. The results showed that even without ISLs, such networks can still beat optical fiber networks for latency.

The routing and wavelength allocation (RWA) problem is a known NP-hard problem in optical networks [22]. In the SON, this problem has also attracted much attention. Considering the dynamic characteristics of satellite constellations, the RWA problem in the SON becomes more complicated. In [11], an ant colony optimization (ACO) with the small window strategy was used to solve the RWA problem in the SON. The inter-satellite distance, link duration, and wavelength idle ratio were introduced as the heuristic functions, and the small window strategy was used to promote the convergence speed. In [12], an ant colony RWA algorithm based on cross-layer design (CL-ACRWA) was proposed. A cross-layer optimization model was built while considering the Doppler wavelength shift, the transmission delay and the wavelength-continuity constraint. Then, an ant colony algorithm was used to find an optimal light path satisfying the above constraints for every connection request. In [13], an extended path-finding RWA algorithm based on ACO (ACO-EP) was proposed. The algorithm can solve the problem of unbalanced wavelength usage. Although swarm intelligence algorithms may give better results, the calculation time is relatively long, and the computational complexity is relatively high. In this paper, to reduce the algorithm complexity, we use Dijkstra's algorithm to address the routing problem and use the wavelength plane mechanism to achieve the

wavelength allocation [1]. In [39], an integer linear programming model (ILP) is proposed to solve satellite-ground optical networks' routing and resource allocation problems. This study aims to minimize the network's total power consumption while ensuring the network's performance and service quality for the users. However, this study does not consider the complexity associated with network size, the dynamic variability of network topology, link duration and switching.

2.2. Satellite assisted survivability technologies

Satellites located in the space area over 400 km high are disaster-independent with the TON. Satellites provide a promising method to recover the interrupted traffic in the TON. In [23], Rafael et al. proposed a method for data evacuation from data centers in disaster-affected regions through software-defined satellite networks. In [24], Hong et al. extended the p-cycle technology to satellite-ground integrated networks, but only used GEO satellites to avoid the instability of satellite-to-ground links caused by the relative motion of the satellites. In [25], Wei et al. proposed path protection in the GEO scenario to solve the problem of satellite-to-ground link interruption caused by cloud covers. In [26], Xin et al. studied the space laser communication (SLC) assisted survivability technologies against shared risk link group (SRLG) failures caused by natural disasters occurring on the ground, proposed the SLC-assisted dedicated protection scheme and the SLC-assisted shared protection scheme, and developed heuristic algorithms to realize those two schemes. That paper only considered using GEO satellites, but GEO has the obvious disadvantage of high propagation delay.

2.3. Our Contributions

This paper researches shared backup path protection against SRLG failures in LEO-satellite-terrestrial integrated optical networks. The contribution of this paper consists of two aspects.

1) Considering the link duration and the number of link switches, an LD-SBPP scheme is proposed in LEO-satellite-terrestrial integrated optical networks.

2) AP-SCRWA algorithm is proposed to pre-calculate the routing and reserve spectrum resources at each continuous snapshot. This algorithm ensures that all required spectrum resources are reserved in advance and a user request will not be blocked due to handovers.

3) Extensive simulations are conducted to evaluate the proposed methods' effectiveness in terms of blocking rate, service performance, and cost. The simulation results demonstrate their effectiveness.

3. Problem Description

This section first presents the integrated architecture of the TON and the SON. Then, the disaster is modeled by the SRLG. Next, the make-before-break (MBB) strategy is elaborated. At last, the proposed LD-SBPP scheme is elaborated in detail.

3.1. The Integrated Architecture of the TON and the SON

Satellites can be divided into three categories according to orbital altitude, i.e., LEO satellite, medium earth orbit (MEO) satellite and GEO satellite. LEO generally refers to orbits with altitudes below 2000 km. Compared with the MEO satellite and the GEO satellite, whose orbital altitude is higher, the LEO satellite has the smallest delay and the lowest path loss. Therefore, in this paper, only LEO satellites are used to form an integrated network with the TON. The satellites are arranged in specific ways to form constellations. The integrated architecture of the TON and the SON is shown in Fig. 1, which is composed of an LEO constellation, a multi-domain TON and satellite-to-ground links.

The TON is divided into two layers, i.e., the primary cross-domain network and the secondary intra-domain network. Each domain in the TON represents a secondary network managed by different carriers and

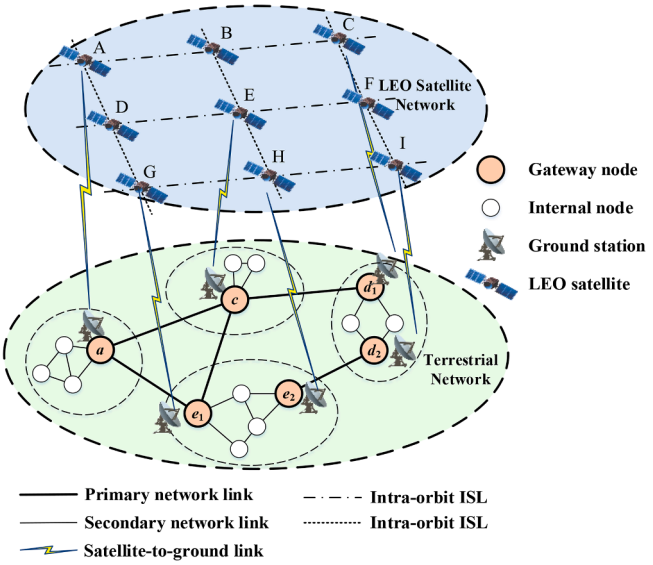


Fig. 1. The LEO-satellite-terrestrial integrated optical network.

even different nations. Domains are interconnected via gateway nodes, and the primary network connects them. Normally, the gateway nodes are better equipped, so we assume that the GSs are deployed at some gateway nodes. An internal node can only get the network information of its domain, while a gateway node can additionally get the information of the primary network and the SON. When a cross-domain request arrives, both ends of the request are first connected to the gateway nodes in their corresponding domains respectively. Then, the two gateway nodes are connected through the primary network. We only study the second process in this paper because the first process does not involve satellite networks. Therefore, the TON is reduced to the primary network, with failures and survivability issues considered. We assume that ISLs and satellite-to-ground links support WDM. Satellites do not allow wavelength conversion due to size and power consumption constraints. In addition, wavelength converters are also only available for the nodes of GSs due to their high cost. However, the deployment of wavelength converters on GSs is necessary. Currently, terrestrial optical networks mainly use dense wavelength division multiplexing (DWDM) technology, in which the wavelength spacing is relatively tight, usually 0.8–2 nm. This tight wavelength spacing is very effective in terrestrial networks. However, optical signals transmitted in space face the challenge of Doppler shift, which can lead to a shift in the signal's wavelength. Therefore, larger wavelength-protected bandwidths are needed to ensure reliable data transmission.

3.2. Disaster Modeling

Natural disasters have a wide range of impacts and usually result in multiple backbone link failures in the TON. On the other hand, for a TON with a large enough coverage area, the impact of a natural disaster will be densely concentrated in one area rather than evenly distributed anywhere in the TON as in ordinary failures. As a result of these two characteristics, a natural disaster may completely cut off the communications of some nodes in the network. Its destructive power is so strong that it renders most of the protection techniques that have been proposed ineffective. In addition, some man-made disasters have the same characteristics as natural disasters, such as power outages. Subsequent analyses include such man-made disasters as well. When a disaster occurs, multiple links in close proximity may fail concurrently. The disaster can be modeled by an SRLG, which is a group of network elements that share a common risk of simultaneous failure [26]. SRLG failures pose a great challenge to network survivability.

As shown in Fig. 2, X is a potential disaster whose impact area covers

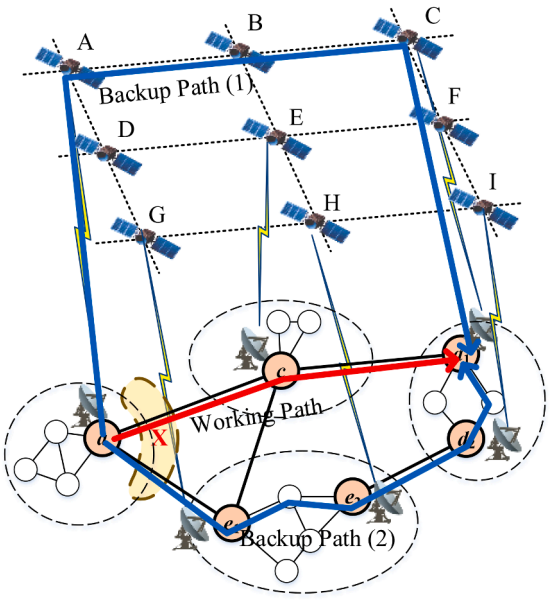


Fig. 2. Satellite assisted path protection.

two links, a-c and a-e1, that form an SRLG. The occurrence of X causes node a to be completely isolated from other terrestrial nodes. The backup path protection relying only on the TON fails because the working and backup paths pass through the same SRLG. This example illustrates the limitation of the survivability technologies, which merely rely on the connectivity of the TON. Improvement can be made with the assistance of satellites since satellites are not vulnerable to terrestrial disasters. The isolated component can be connected by satellite-to-ground links, ISLs, and terrestrial links. In such a satellite-terrestrial integrated network, protection technologies such as backup path protection can be applied. With the assistance of the SON, protection in disaster situations is possible. The SBPP scheme will be adopted, because it is suitable for dynamic satellite networks, and the handover process is simple to operate. Previous research has confirmed that SBPP has higher spectral efficiency than dedicated backup path protection (DBPP) [1]. In the SBPP scheme, a pair of working lightpaths and protection lightpaths will be established for each request. In the case of link failure, SBPP specifies that the backup resources of two requests whose working paths are link-disjoint can be shared with each other. In [25], the authors gave the first four regulations for satellite-assisted protection schemes in SRLG failure conditions as follows.

(1) The working lightpath and the backup lightpath of a request must be link-disjoint and SRLG-disjoint. In other words, the two lightpaths cannot pass through a common link nor can they pass through a common SRLG.

(2) Since there are more spectrum resources in the TON and the cost of spectrum resources in the TON is much lower than in space, the working lightpath of a request is only allowed to be established in the TON.

(3) The backup lightpath of a request should avoid using the resources in the SON whenever possible, which is also due to cost and delay considerations.

(4) The resources should be shared as much as possible between the backup lightpaths of different requests, thus improving resource efficiency. Two backup lightpaths of two different requests are allowed to share backup resources on the common link they pass through only if their working lightpaths are link-disjoint and SRLG-disjoint. This principle ensures that after a failure, any disconnected requests can be recovered via the backup path without spectrum clash.

(5) When establishing a space lightpath, the space laser link between two satellites or between a satellite and a GS with a long link duration should be used as much as possible.

This paper still complies with the first four regulations. Since the space laser link between two LEO satellites remains stable and available only for a specific period of time, link-switching is inevitable. Frequent link-switching can cause the fluctuation of QoT. We give the fifth regulation to minimize the impact of link-switching.

3.3. The Make-Before-Break (MBB) Strategy

Due to the handover between two LEO satellites or between a LEO satellite and a GS, an end-to-end backup lightpath may not sustain until the deadline of the user request. There may be multiple backup lightpaths continuous in time for a user request.

As presented in Fig. 3(a), the working lightpath is $a \rightarrow c \rightarrow d_1$ and the backup lightpath is $a \rightarrow A \rightarrow B \rightarrow C \rightarrow d_1$. As the satellite moves, LEO satellite A will be invisible for GS a. At that time, the backup lightpath $a \rightarrow A \rightarrow B \rightarrow C \rightarrow d_1$ will be interrupted. To ensure the required survivability requirement, a new backup lightpath should be established. Due to the process of ATP, optical terminals generally take almost 60 s to complete the ATP process [16]. The make-before-break (MBB) strategy is adopted to avoid the data loss caused by the handover. The make-before-break strategy means that a new path is established before the original path breaks. In [27], the authors focused on the MBB strategy in the OTN layer. They proposed an exact and scalable optimization model for computing a minimum bandwidth rerouting scheme subject to the MBB strategy. In [28], a non-disruptive bandwidth defragmentation heuristic with an MBB strategy was proposed for optical networks. In [29], the MBB strategy was used to ensure seamless handover for network mobility. In the process of the handover in the SON, the MBB strategy can avoid data loss. In Fig. 3 (b), LEO satellite A is about to become invisible to GS a and LEO satellite G has become visible to GS a.

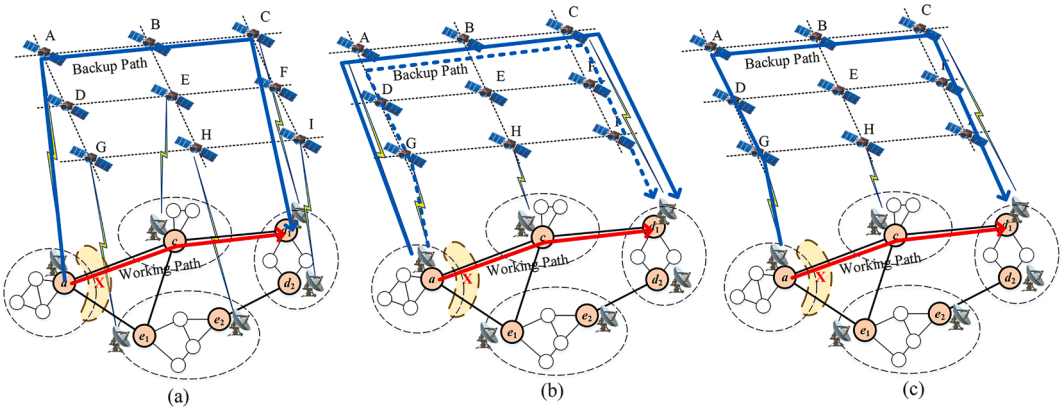


Fig. 3. The backup lightpath handover in the SON (a) before handover, (b) the make-before-break strategy, (c) after handover.

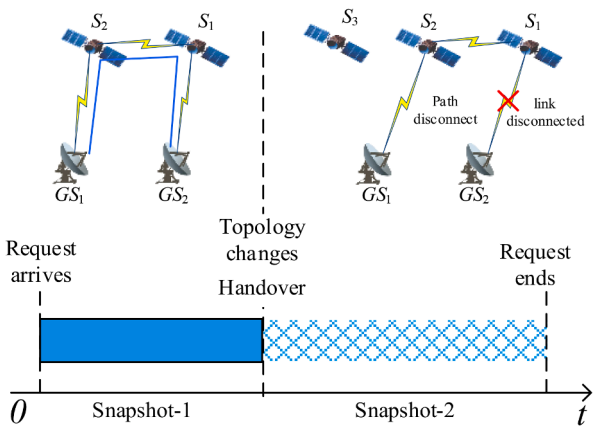


Fig. 4. The handover between continuous snapshots.

At this time, a new backup lightpath $a \rightarrow G \rightarrow D \rightarrow A \rightarrow B \rightarrow C \rightarrow d_1$ is established. Then, all occupied wavelength resources along the lightpath $A \rightarrow B \rightarrow C \rightarrow d_1$ are released and the working optical terminals at GS a and LEO satellite A which form the satellite-to-ground link $a \rightarrow A$ enter an idle state. As presented in Fig. 3 (c), lightpath $a \rightarrow G \rightarrow D \rightarrow A \rightarrow B \rightarrow C \rightarrow d_1$ is regarded as the backup lightpath. The MBB strategy requires all involved GSs and LEO satellites to have at least an idle optical terminal. If there does not exist an idle optical terminal when the traffic load is high, data loss is inevitable in the process of the handover. The proposed LD-SBPP scheme uses the space laser links with long duration as much as possible to avoid too many handovers.

The SDN framework is adopted to realize the MBB strategy. In [30], the SDN was introduced into the space–air–ground integrated network (SAGIN) and a hierarchical domain-based SDN-enabled SAGIN architecture was designed. In [31], an intelligent spectrum management framework based on SDN and artificial intelligence (AI) was proposed. The SDN transforms the heterogeneous satellite and terrestrial networks into an integrated satellite and terrestrial network (ISTN) with reconfigurability and interoperability. In [32], an optimal controller placement and satellite-to-controller assignment which minimizes the average flow setup time with respect to varying traffic demands was presented. In [33], a handover method was proposed for network mobility that utilizes the multi-homing feature of a mobile router in SDN-based network mobility. The maintenance process of the SDN during the handover can be summarized as follows. A SDN controller is placed at a LEO satellite or a GEO/MEO satellite. All backup lightpaths are established by this SDN controller. The routing module in this SDN controller pre-calculates the routing in the SON for each backup lightpath based on the continuous snapshot. We take the handover between a LEO satellite and a GS as an example. The SDN controller monitors the durations of all optical satellite-to-ground links at all times. When it finds a LEO satellite is about to become invisible for the connected GS, it determines a visible LEO satellite for handover according to the original arrangement. It sends the control message to the GS and the selected LEO satellite to conduct the ATP process. After this new optical satellite-to-ground link is successfully established, it sends the control message to the GS and the LEO satellite which is about to become invisible to release the working optical terminals. At this time, the handover is successfully processed.

3.4. The LD-SBPP Scheme

In order to handle the dynamic satellite topology, time is discretized, and the smallest time unit is defined as the time-slot t_s. All subsequent time refers to the time after normalization with t_s. The dynamic satellite topology can be regarded as static in several continuous time-slots, and these continuous time-slots constitute a snapshot. In general, the duration of each snapshot is different, depending on the position relationship between two satellites and between a satellite and a GS. The start time and duration of snapshots are obtained according to the ephemeris. Between two snapshots, the process of link update is defined as handover. If the duration of a request spans multiple snapshots with different topologies, it may need to perform the link-switching at the time of handover. The conventional approach is to reroute disconnected connections at the moment of handover. If rerouting fails, the request is discarded.

In Fig. 4, the request r arrives at the first snapshot and ends at the second snapshot. The topology changes when the second snapshot arrives, causing the original path to disconnect. The rerouting may fail due to a lack of sufficient resources. If so, the tail of r (the grid part) is discarded. This situation is intolerable for the survivability technology. Pre-calculated switching can solve this problem by calculating the rerouting process in advance when the request arrives. If the subsequent snapshots do not have enough resources, the pre-calculated switching process will reject the request when it arrives. Those requests that are not rejected will never be discarded midway. At each continuous snapshot, the routing is pre-calculated and spectrum resources are reserved in advance. At the end of the last time-slot of a snapshot, a handover needs

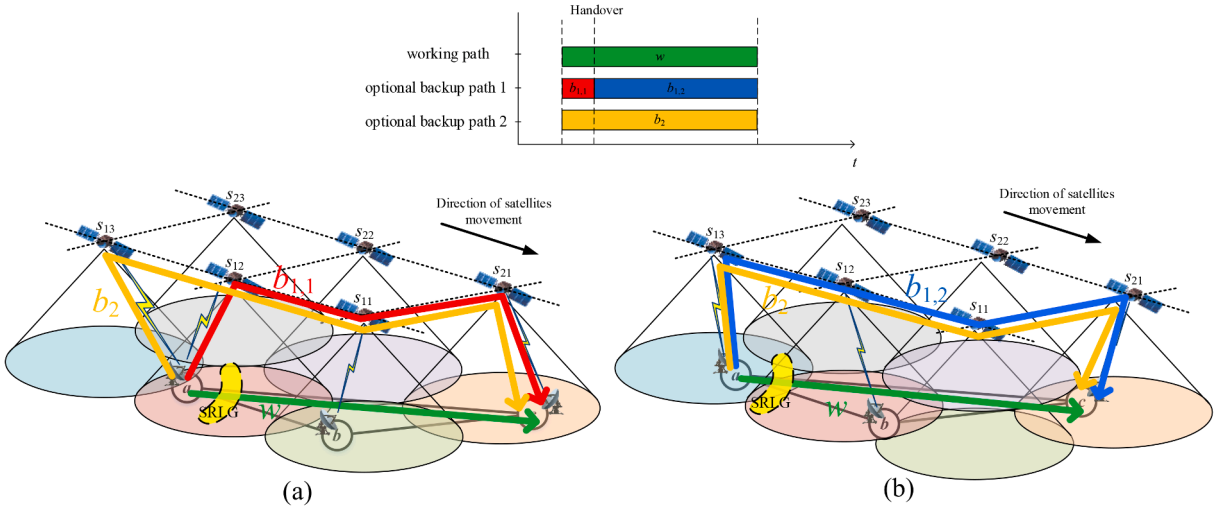


Fig. 5. Effect of link duration on the link-switching (a) backup lightpaths in snapshot-1 and (b) backup lightpath in snapshot-2.

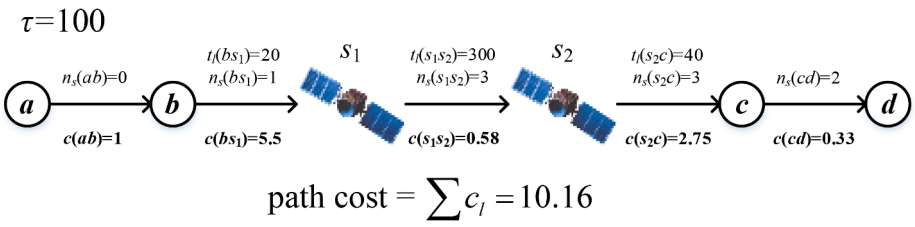


Fig. 6. Example of path cost calculation.

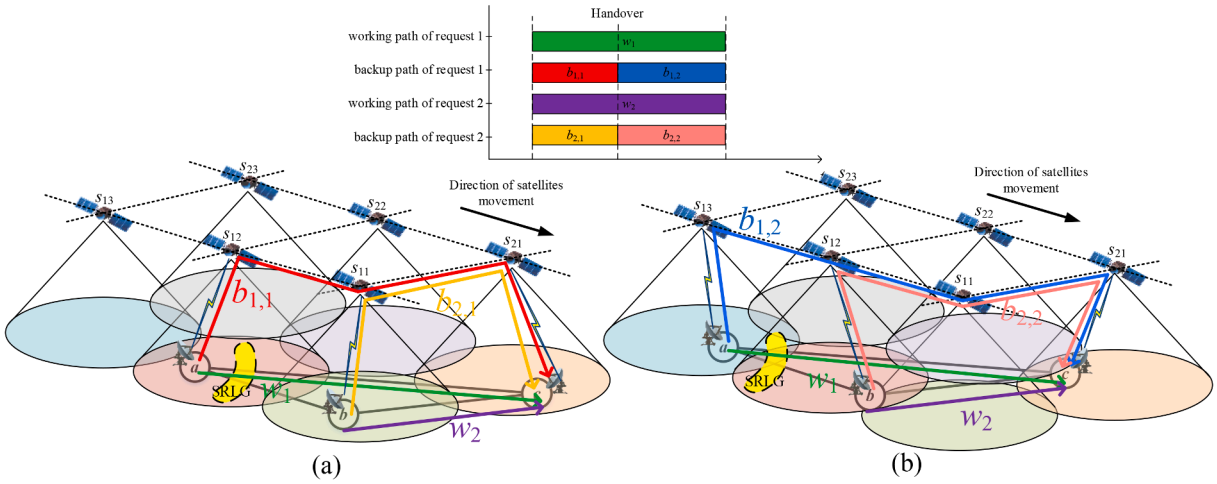


Fig. 7. Example of LEO backup path handover (a) backup lightpaths in snapshot-1 and (b) backup lightpath in snapshot-2.

to be performed. There are three steps to a handover. The first step is to update the network topology. The second step is to complete the switching of all paths based on the pre-calculated plan. The third step is to delete the link-switching plan that has been executed and deliver the remaining plans to the next snapshot. The route composed of long-duration links should be selected as much as possible to avoid frequent link-switching. The LD-SBPP scheme defines a cost for each link. The cost of the terrestrial link in the LD-SBPP scheme is defined as:

$$c_l = \begin{cases} \dfrac{1}{(1+n_s)} & \text{if it is free or shareable} \\ \infty & \text{if it is unshareable} \end{cases} \tag{1}$$

where n_s denotes the number of requests shared on the wavelength w of link l at time t. The cost of the space laser link in the LD-SBPP scheme is defined as:

$$c_l = \begin{cases} \dfrac{1}{(1+n_s)} + \dfrac{\tau}{t_l} & \text{if it is free or shareable} \\ \infty & \text{if it is unshareable} \end{cases} \tag{2}$$

where t_l indicates the link duration of l obtained from the ephemeris. If the link duration exceeds the request duration t_h, let $t_l = t_h$, since the excess link duration is not necessary. τ is a constant, which normalizes $1/t_l$. By varying the value of τ, t_l can be made to have different extents of impact on c_l. The cost c_l is the sum of two terms. The first term $1/(1+n_s)$ indicates the sharing state and the second term τ/t_l indicates the link duration. The first term encourages sharing. The second term encourages the selection of space laser links that have a long duration rather than those that will soon fade.

Fig. 5 gives an example of the effect of the link duration on the link-switching. For request a-c, there are two optional backup paths $b_{1,1}$ and b_2. Because of the movement of satellites, GS a is about to leave the coverage of satellite s_{12}, that is to say, link a-s_{12} is about to expire. Once it leaves, $b_{1,1}$ will have to switch to $b_{1,2}$, while b_2 does not have this issue, as shown in Fig. 5(b). If the link a-s_{12} with too short duration is not chosen when the request arrives, but the path b_2 is adopted, the link-switching is not necessary. The second term avoids using those short-duration links by increasing the cost of them. Although this might lead to a path with more hops, it is worthwhile to reduce the number of link-switching. The LD-SBPP scheme will choose the path with the lowest cost.

The cost of a path is the sum of the link costs along the path. As shown in Fig. 6, the costs of the terrestrial links (a-b, c-d) are calculated by Eq. 1, and the costs of the space laser links (b-s_1, s_1-s_2, s_2-c) are calculated by Eq. 2, after which they are summed to obtain the total costs of the path.

A general example of the proposed LD-SBPP scheme is given in Fig. 7. Assuming a request goes from node a to node c, and the SRLG is shown in the yellow area. Constrained by network connectivity, it is not possible to establish a pair of SRLG-disjoint working path and backup path in TON. So, the backup path needs to use the LEO satellite network. As shown in Fig. 7(a), the backup path is (a-s_{12}-s_{11}-s_{21}-c). Similarly, the

Table 1
List of Notations.

Notation	Description
i	The index of snapshots.
$G(V,E)_i$	The satellite-terrestrial integrated network in the i-th snapshot.
V_i	The set of nodes in G in the i-th snapshot.
E_i	The set of links in G in the i-th snapshot.
$G_T(V_T, E_T)$	The terrestrial part of G.
V_T	The set of nodes in G_T.
E_T	The set of links in G_T.
$G_S(V_S, E_S)_i$	The satellite part of G (also including GSs and satellite-to-ground links) in the i-th snapshot.
V_{Si}	The set of nodes in G_S in the i-th snapshot.
E_{Si}	The set of links in G_S in the i-th snapshot.
l	Link index.
w	Wavelength index.
t	Time index.
c_l	The cost of link l.
t_l	The link duration of l.
a	The parameter used to represent the link type, for links in E_T, $a = 1$, for links in E_s, a is a large number.
τ	The parameter used to normalize t_l.
n_s	The number of requests sharing resources.
n_r	The number of link-switching.
r	User request.
s	The starting node of the request.
d	The destination node of the request.
t_a	The arriving time of the request.
t_h	The duration of the request.
i_r	The snapshot index when r arrives.
M	The number of subsequent snapshots that request spans.
m	The index of subsequent snapshots that request spans.
B	The set of paths established for the request, including the paths of all snapshots in the request duration.
b_m	An Element in B, the path of the m-th snapshot in the request duration.
w_{tm}	The index of the terrestrial wavelength assigned to the request in the m-th snapshot in the request duration.
w_{sm}	The index of the satellite wavelength assigned to the request in the m-th snapshot in the request duration.
W_t	The set of w_{tm}.
W_s	The set of w_{sm}.
S	Maximum number of link-switching allowed for each request.

backup path for another request from node b to node c is b-s_{11}-s_{21}-c. Because the working paths of these two requests are disjoint, they can share backup resources. Specifically, both backup paths pass through links s_{11}-s_{21} and s_{21}-c, so the two backup paths can share the same wavelength resources on the two links, thus reducing the total resource consumption. However, due to the movement of the satellites, the backup path does not remain permanent for the whole request duration. Therefore, each of the backup paths needs to be switched. The good thing is that the ephemeris is predictable, so resources can be planned and reserved in advance for requests to avoid resource conflicts. According to the ephemeris, node a will leave the coverage of satellite s_{12} and enter the coverage of s_{13}, so the backup path of request 1 will be switched to a-s_{13}-s_{12}-s_{11}-s_{21}-c, and the backup path of request 2 will be switched to b-s_{12}-s_{11}-s_{21}-c, as shown in Fig. 7(b). The two requests can still share resources on the common links that the two backup paths pass through (links s_{12}-s_{11}, s_{11}-s_{21}, and s_{21}-c). For a request, the backup path is different before and after the handover, but the working path is unchanged.

4. The P-SCRWA algorithm

The P-SCRWA algorithm establishes a pair of working and a set of backup paths for each request as it arrives. The set of backup paths contains backup paths within all continuous snapshots during the request duration. First of all, the snapshots are divided according to the movement of the satellites. Whenever the topology of the satellite network changes (at least one link in the satellite network changes), a new snapshot is generated. Store the start time, duration and network topology of each snapshot in the ephemeris for the P-SCRWA process. The topology of the i-th snapshot is $G(V,E)_i$. The satellite part of G (also including GSs and satellite-to-ground links) is $G_S(V_S,E_S)_i$, and the terrestrial part of G is $G_T(V_T,E_T)$. Note that since G_T does not change with the snapshot, it is not indexed with i. When a request arrives, the P-SCRWA process is started and a pair of working and backup paths is established for the request. For the working path, only the TON can be used, so the impact of handover does not need to be considered. While for the backup path, the P-SCRWA algorithm for LD-SBPP ensures that the request will not be interrupted due to handover by calculating the paths of a request in all snapshots within its duration and allocating resources when the request arrives. Specifically, first calculate the path in the current snapshot, and then judge whether this path is still available in the next snapshot within the request duration. If available, keep it, otherwise calculate the new available path, and so on. When calculating the backup path within a snapshot, this scheme realizes wavelength assignment by traversing the combination of all satellite wavelengths and terrestrial wavelengths. Two wavelength planes [34] are constructed to represent the satellite and terrestrial available link topology at a specific wavelength. The path with the minimum cost is found in the wavelength planes. Table 1 lists all used notations in the developed algorithms.

The P-SCRWA algorithm for the working path finds the shortest path among the free resources in all wavelength planes [34]. Because this algorithm is relatively simple, it is omitted here. The P-SCRWA algorithm for the backup path is shown below. The inputs of **Algorithm 1** include the request r, the maximum number of switches allowed S, the ephemeris and the snapshot index when r arrives i_r. r has four attributes, i.e., the pair of end nodes s and d, the starting time t_a, the duration t_h. The output includes the set of backup paths for all snapshots of request r during its duration, namely B, as well as the terrestrial and satellite wavelengths assigned to each path. When r arrives, first calculate the number of snapshots that r spans. Then establish a backup path for r in the current snapshot (Line 3). Lines 4–17 establish backup paths for subsequent snapshots. Before calculating the path, first verify whether the path of the previous snapshot is still available (Line 5), and if so, it does not need to calculate a new path and the previous path is preserved (Lines 6–9), otherwise r needs a new path and a switch. The number of link-switching of r is limited to no more than S. If it exceeds S, r will be rejected (Lines 11–14), because the very-long-last requests that require too many switches will greatly increase the complexity, and such requests are not suitable for this LEO protection scheme. Finally, the set of backup paths for all snapshots within the duration of r is obtained. If B is successfully found, the resources on b_0 will be occupied at the current time, and the other paths in B will be added to the corresponding handover plans, and the link state information at the time of the next M handovers will be updated.

Algorithm 1: The P-SCRWA for backup path

Input: Request $r(s,d,t_a,t_h)$, the maximum number of link-switching allowed S, ephemeris within the duration of r, the snapshot index when r arrives i_r;

Output: The set of backup paths of snapshots B, the set of terrestrial wavelengths assigned W_t, the set of satellite wavelengths assigned W_s;

Set n_r=0, m=0;
Calculate the number of subsequent snapshots that r spans, which is M, by the ephemeris;
Establish a backup path for r in the current snapshot by **Algorithm 2**;
for m=1,2,…,M **do**
    Determine if the path of the previous snapshot b_{m-1} is still available by **Algorithm 3**;
    if available **then**
        b_m=b_{m-1};
        w_{tm}=w_{tm-1};
        w_{sm}= w_{sm-1};
    else
        n_r=n_r+1;
        if n_r>S **then**
            reject r;
        end if
        Establish a backup path for r in m-th snapshot by **Algorithm 2**;
    end if
end for

The RWA in each snapshot is done by **Algorithm 2**. **Algorithm 2** first eliminates the unavailable links (Lines 1–5) and then finds a path with minimum cost (Lines 11–24). If no available path can be found in any one of the snapshots, the request will be rejected (Lines 25–27).

Algorithm 2: Backup path RWA in *m*-th snapshot

Input: Request $r(s,d,t_a,t_h)$, integrated network $G(V,E)_{i_r+m}$;

Output: The m-th backup path of snapshots b_m, terrestrial wavelength assigned w_{tm}, satellite wavelength assigned w_{sm};

1: **for** each link l in E_T **do**
2: **if** the working path of r traverses l or it traverses the SRLG containing l **then**
3: Mark l as unavailable;
4: **end if**
5: **end for**
6: **if** m=0 **then**
7: t=current time-slot;
8: **else**
9: t=the first time-slot of the m-th snapshot;
10: **end if**
11: **for** each available link l in E **do**
12: **for** each wavelength w **do**
13: Calculate cost c_l by Equation (1) and (2) for link l with wavelength w at time t, and if the link l with the wavelength w is not available (be occupied by working path or not shareable), c_l=∞;
14: **end for**
15: **end for**
16: **for** each combination of w_t and w_s **do**
17: Construct virtual wavelength plane with G_T of w_t and G_S of w_s;
18: Obtain a path p with the lowest cost by Dijkstra's algorithm;
19: **if** b_m=null **or** the cost of p is lower than that of b_m **then**
20: b_m=p;
21: w_{tm}=w_t;
22: w_{sm}=w_s;
23: **end if**
24: **end for**
25: **if** b_m=null **then**
26: Reject r;
27: **end if**

Algorithm 3 is used to verify that the path of the previous snapshot is still available.

Algorithm 3: Path availability verification

Input: Path of the previous snapshot $b_{m\text{-}1}$ with the wavelength assigned $w_{tm\text{-}1}$ and $w_{sm\text{-}1}$;
Output: Whether $b_{m\text{-}1}$ is still available in the m-th snapshot;
1: **for** each link l in $b_{m\text{-}1}$ **do**
2: **if** $l \in E_T$ **then**
3: $w = w_{tm\text{-}1}$;
4: **else**
5: **if** $l \notin E_{S_{i,}+m}$ (l is not an existing path in m-th snapshot) **then**
6: **return** NOT available;
7: **else**
8: $w = w_{sm\text{-}1}$;
9: **end if**
10: **end if**
11: **if** the wavelength w in link l is NOT free or NOT sharable **then**
12: **return** NOT available;
13: **end if**
14: **end for**
15: **return** available;

The main part of **Algorithm 2** contains two loops. The complexity of the loop of lines 11–15 is $O(|W_T|*|E|)$, where $|W_T|$ is the total number of wavelengths in each fiber link and $|E|$ is the total number of links. The complexity of the loop of lines 16–24 is $O(|W_S|*|W_T|*|V|^2)$, where $|W_S|$ is the total number of wavelengths in each satellite link and $|V|$ is the number of network nodes. So, the total computational complexity of Algorithm 2 is $O(|W_T|*(|E|+|W_S|*|V|^2))$. The complexity of **Algorithm 3** is $O(|b|)$, where $|b|$ denotes the number of links contained in the path. Since the complexity of **Algorithm 3** is much smaller than that of **Algorithm 2**, it can be ignored. **Algorithm 1** loops **Algorithm 2** at most $1 + M$ times. So, the total computational complexity of **Algorithm 1** is $O(M*|W_T|*(|E|+|W_S|*|V|^2))$, where M is determined by the duration.

5. Simulations

To evaluate the performance of the proposed algorithm, we constructed a satellite-terrestrial integrated network. The LEO constellation uses Iridium-like orbital parameters. It consists of 6 near-circular orbits at an altitude of 780 km, with 11 satellites in each orbit. The orbit inclination is 86.4 degrees. The phase disparity between two adjacent orbits is 16.4 (180/11) degrees, which means that the adjacent two satellites in different orbits are staggered by 16.4 degrees. Co-rotating orbits are spaced 31.6 degrees apart and counter-rotating orbits (one and six) are spaced 22 degrees apart [35]. The minimum elevation of GSs is 8.2 degrees [14]. This design ensures maximum coverage. Each satellite in the constellation can establish two intra-orbit ISLs with the two adjacent satellites in the same orbit and two inter-orbit ISLs with the two adjacent satellites in the adjacent two orbits. The exception is the connections cannot be established between counter-rotating orbits, and the inter-orbit connections are temporarily closed when the satellites move to high latitudes (60–90 degrees). All the GSs in a satellite's coverage area can connect with the satellite if the elevation is more significant than the minimum elevation. It is assumed that there are 16 wavelengths in each ISL and satellite-to-ground link [36]. To explore the relationship between the number of different interstellar or star-earth wavelengths and the blocking rate, the interstellar and star-earth link wavelengths are set to {12,16,20,24}. The TON consists of 11 Pacific Rim gateway nodes, six of which have GSs deployed, and the submarine links between them. It is assumed that there are 80 wavelengths in each terrestrial link [37]. Each node is equipped with a GS to connect with satellites. The information on the gateway nodes deployed with the GSs is shown in Table 2. The connection relationship of these nodes is shown in Fig. 8, with red circles representing nodes deployed with GSs and black circles representing other nodes. The SRLGs simulating disasters caused by geological movements of the seafloor are circled by the dashed lines. Note that the figure only represents the link connection topology, not the relationship between the nodes' location coordinates.

The requests arrive according to a Poisson process and the duration follows a negative exponential distribution. The satellite network is

Table 2
Information of terrestrial nodes.

Symbol	Coordinate (Longitude, Latitude)	Name
a	(31°N, 121°E)	Shanghai
b	(22°N, 114°E)	Hong Kong
c	(36°N, 140°E)	Tokyo
d	(34°N, 118°W)	Los Angeles
e	(33°S, 151°E)	Sydney
f	(1°N, 104°E)	Singapore

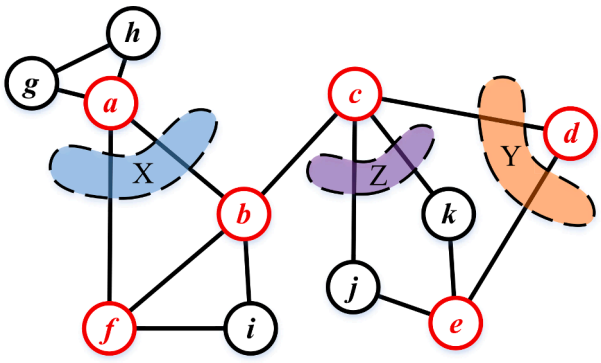


Fig. 8. The test TON.

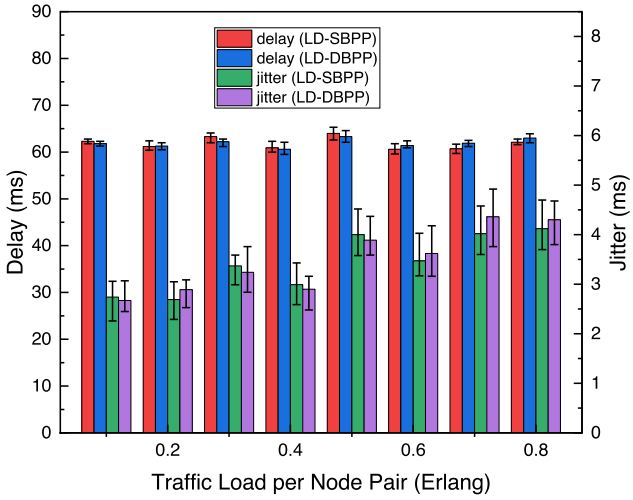


Fig. 9. End-to-End Delay and Delay Jitter.

simulated for about 400 min. The orbit period of the adopted Iridium constellation is about 100 min. The minimum duration of all snapshots is about 7 s and the maximum duration of all snapshots is about 97 s. The developed P-SCRWA algorithm pre-calculates the routing and reserves spectrum resources at each continuous snapshot for each user request. In order to verify the P-SCRWA algorithm with moderate snapshots, the mean request duration is set to 200 s. The values of the mean request interval time are determined by the traffic load. When the traffic load is 0.1 Erlang, the mean request interval time is 36 s. When the traffic load is 0.8 Erlang, the mean request interval time is 4.5 s. The two end nodes of each request are both terrestrial nodes (gateway nodes or non-gateway nodes). Requests whose working path or backup path cannot be established will be rejected. Due to the movement of satellites, the satellite serving the ground source terminal or destination terminal may be taken over by another satellite during the communication process, called handover. A handover changes the previous routing path, and more handovers mean more path changes during the routing process, which is detrimental to the stability of end-to-end delay and the implementation of routing protocols [38]. The maximum number of link-switching allowed for each request S is set to 3.

5.1. End-to-End Delay and Delay Jitter

Due to the handover between two satellites or between a satellite and a GS, there may be multiple backup lightpaths continuous in time for a user request. Assuming $\{LP_1, LP_2, \ldots, LP_N\}$ are multiple backup lightpaths of a user request and $\{DT_1, DT_2, \ldots, DT_N\}$ are the durations of those lightpaths. Then, the end-to-end delay is calculated as $(DT_1{*}length(LP_1)/c + DT_2{*}length(LP_2)/c + \ldots + DT_N{*}length(LP_N)/c)/(DT_1 + DT_2 + \ldots + DT_N)$. The function *length*() represents the average length of a lightpath and notation c denotes the speed of light.

The delay jitter refers to the delay difference between two backup lightpaths. The average maximum delay jitter of all user requests is calculated. Fig. 9 shows that the average end-to-end delay is about 61.9 ms and the average maximum delay jitter is about 3.42 ms for the proposed LD-SBPP scheme.

5.2. Blocking Probability with Short and Long durations

For use requests with the mean request duration of 200 s, the blocking probability of the protection schemes is shown with different traffic loads. In addition to the LD-SBPP scheme, we also evaluate the LD-DBPP scheme as well as the two schemes with no satellite assistance (only TON). The results are shown in Fig. 10, where the blocking probability of the LD-SBPP and LD-DBPP schemes increases with the traffic load. There is no difference in the blocking probability between the two TON schemes, which is attributed to the fact that the terrestrial wavelength resources are sufficient and the sharing of terrestrial resources does not reduce the blocking probability. The average blocking probabilities of the LD-SBPP and the LD-DBPP schemes are 95.6 % and 91.5 % lower than those of the TON schemes, respectively. This

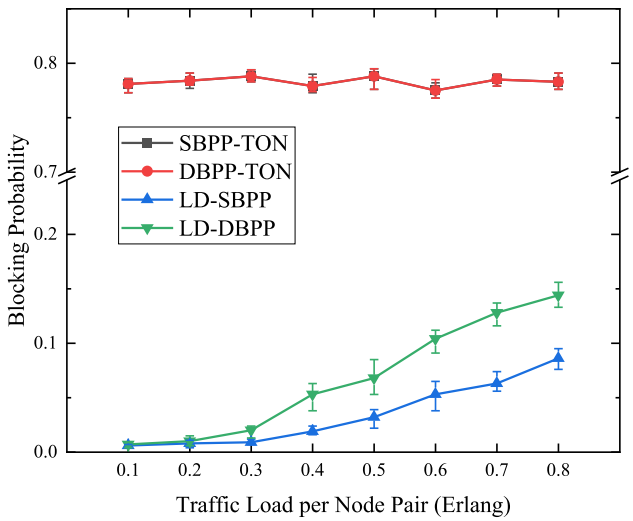


Fig. 10. Blocking probability with short duration.

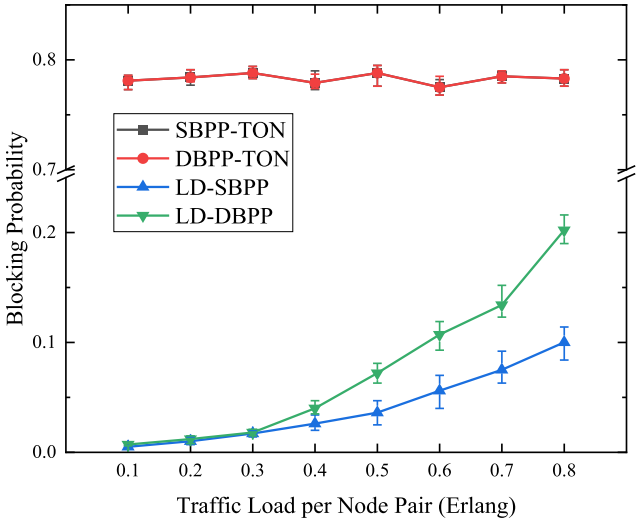


Fig. 11. Blocking probability with long duration.

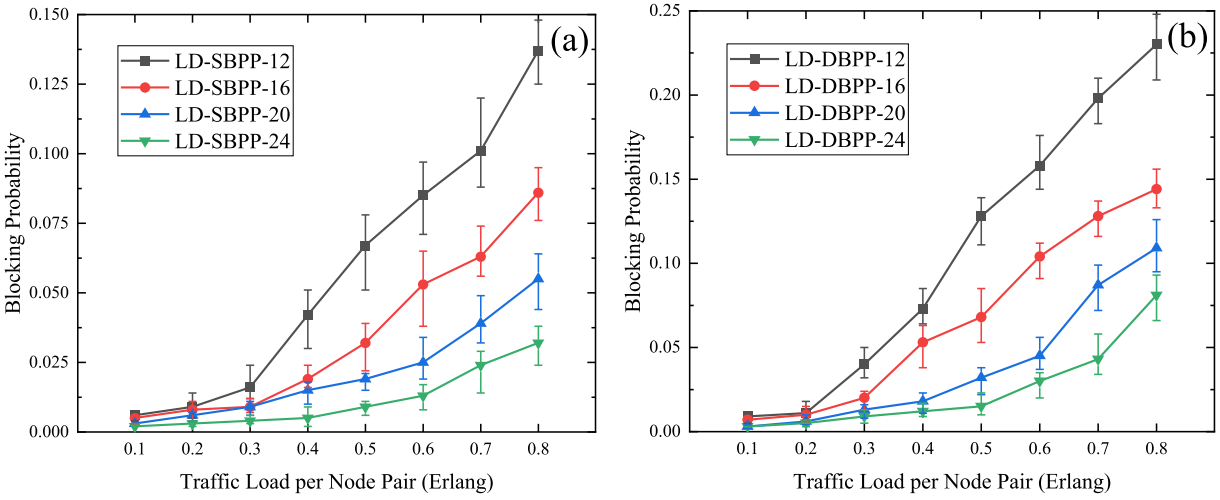


Fig. 12. Blocking probability with satellite capacity variation: (a) the LD-SBPP scheme and (b) the LD-DBPP scheme.

indicates that relying only on the TON is ineffective in the case of SRLG failure, while the LEO satellite network can effectively complement the TON. The LD-SBPP scheme has a 0.032 lower average blocking probability than the LD-DBPP scheme, which corresponds to a 48.3 % reduction. This demonstrates that sharing can improve the utilization of wavelength resources.

Fig. 11 presents the blocking probability for use requests with the mean request duration of 400 s. The long duration will increase the blocking probability. The blocking probability of the LD-DBPP scheme increases by about 10.9 % and the blocking probability of the LD-SBPP scheme increases by about 17 %. At this time, the blocking probability of the LD-DBPP scheme is 7.4 % and the blocking probability of the LD-SBPP scheme is 0.4 %. The blocking probability of the LD-SBPP scheme is also much smaller than that of the LD-DBPP scheme.

5.3. Impact of Space Laser Link Capacity on Blocking Probability

The SON has fewer wavelength resources than the TON, so it is considered more important. Increasing the number of wavelengths in each SON link is supposed to result in a significant reduction in the blocking probability and vice versa.

We set the number of wavelengths of the SON to 12, 16, 20, and 24, and the results are shown in Fig. 12. The blocking probability of both LD-SBPP and LD-DBPP schemes decreases as the number of wavelength

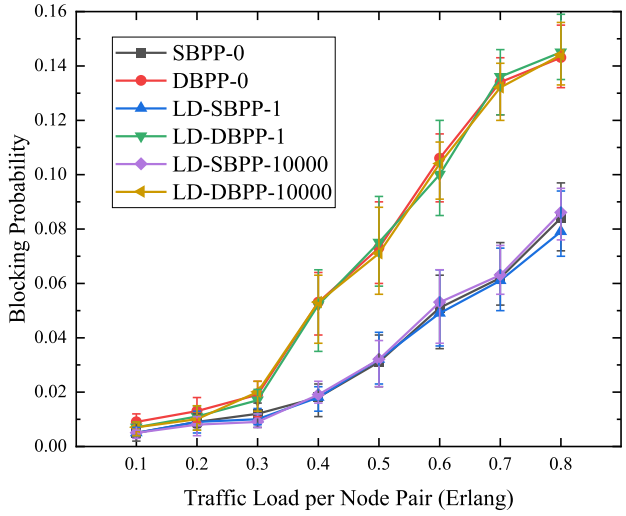


Fig. 13. Blocking probability with different value of τ.

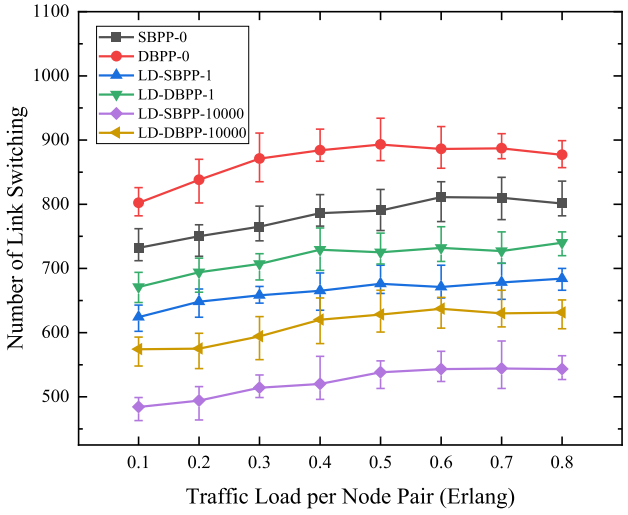


Fig. 14. Number of link-switching with different value of τ.

resources increases. Specifically, the average blocking probability of the 12-wavelength LD-SBPP, the 16-wavelength LD-SBPP, the 20-wavelength LD-SBPP, and the 24-wavelength LD-SBPP schemes are 0.058, 0.035, 0.021, and 0.012, respectively. Taking the 12-wavelength LD-SBPP scheme as the benchmark, the average blocking probability of the 16-wavelength LD-SBPP, the 20-wavelength LD-SBPP, and the 24-wavelength LD-SBPP schemes are 40.4 %, 63.1 %, and 80.1 %, lower than it respectively. The average blocking probability of the 12-wavelength LD-DBPP, the 16-wavelength LD-DBPP, the 20-wavelength LD-DBPP, and the 24-wavelength LD-DBPP schemes are 0.106, 0.067, 0.039, and 0.025, respectively. Taking the 12-wavelength LD-DBPP scheme as the benchmark, the average blocking probability of the 16-wavelength LD-DBPP, the 20-wavelength LD-DBPP, and the 24-wavelength DBPP schemes are 37.0 %, 63.0 %, and 76.6 % lower than it respectively. These results illustrate that the SON wavelength resources have a strong impact on the blocking probability and the availability of these schemes is conditional on adequate satellite wavelength resources.

5.4. Effect of Different τ on the Number of Link-Switching and Blocking Probability

The parameter τ in Eq. 1 affects the preference of the algorithm when routing. Specifically, the larger τ, the more the algorithm prefers s to

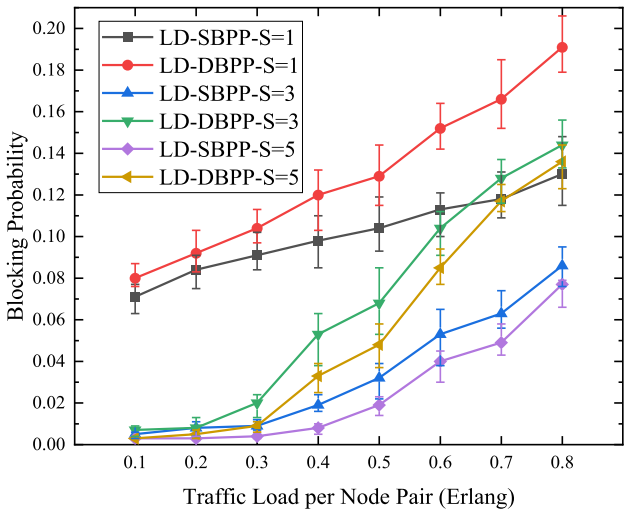


Fig. 15. Blocking probability with different value of *S*.

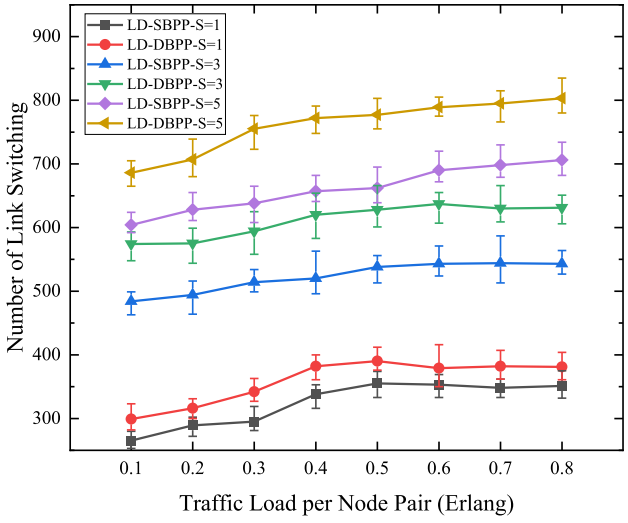


Fig. 16. Number of link-switching with different value of *S*.

select a path with longer durations, resulting in fewer switches, and vice versa. A smaller number of switches are beneficial in reducing the blocking probability. However, a larger τ may lead to the selection of a path with a higher number of hops by over-considering the duration when routing, which in turn will increase the blocking probability.

The test results are shown in Fig. 13 and Fig. 14. Three values of τ were tested, that is, 0 (i.e., no consideration of link duration at all), 1 and 10000. In Fig. 13, there is almost no difference in the blocking probability results for different values of τ. In other words, the value of τ has almost no effect on the blocking probability. This might result from the fact that the positive and negative effects of τ on the blocking probability exactly cancel out in the current test conditions. In Fig. 14, the number of link-switching increases with decreasing τ. For the LD-SBPP scheme, the number of link-switching in the case of $\tau = 10000$ is 33.1 % less than that of $\tau = 0$, and 15.1 % less than that of $\tau = 1$. For the LD-DBPP scheme, the number of link-switching in the case of $\tau = 10000$ is 29.5 % less than that of $\tau = 0$, and 17.5 % less than that of $\tau = 1$. One link-switching results in one additional route calculation, so the number of link-switching reflects the complexity. Combining the two factors of blocking probability and link-switching, a larger value of τ is a better choice. The conclusion is that link duration should be given priority and resource sharing should be considered in links of comparable duration.

5.5. Blocking Probability with Different Maximum number of link-switching allowed

The maximum number of allowed link-switching S is set to reject requests that last too long, which are not suitable for protection with LEO. A larger S leads to higher complexity, while a smaller S leads to rejecting too many requests, making the blocking probability higher. Therefore, both considerations need to be taken into account when selecting S.

We verified the blocking probability for the three cases of $S = 1$, $S = 3$, and $S = 5$. The results are shown in Fig. 15 and Fig. 16. In Fig. 15, the blocking probability decreases with increasing S. For the LD-SBPP scheme, the average blocking probability for the $S = 5$ condition is 26.4 % lower than that for $S = 3$, and 74.9 % lower than that for $S = 1$. For the LD-DBPP scheme, the average blocking probability for the $S = 5$ condition is 18.4 % lower than that for $S = 3$, and 57.8 % lower than that for $S = 1$. Notice that the difference between the cases is more pronounced when the traffic load is low, and becomes smaller when the load is high. This is due to the fact that at higher traffic load, link wavelength capacity becomes the main factor affecting the blocking probability, and the effect of the limitation on the number of link-switching is diminished. In Fig. 16, for the LD-SBPP scheme, the average number of link-switching for the $S = 5$ condition is 26.4 % higher than that for $S = 3$, and 103.7 % higher than that for $S = 1$. For

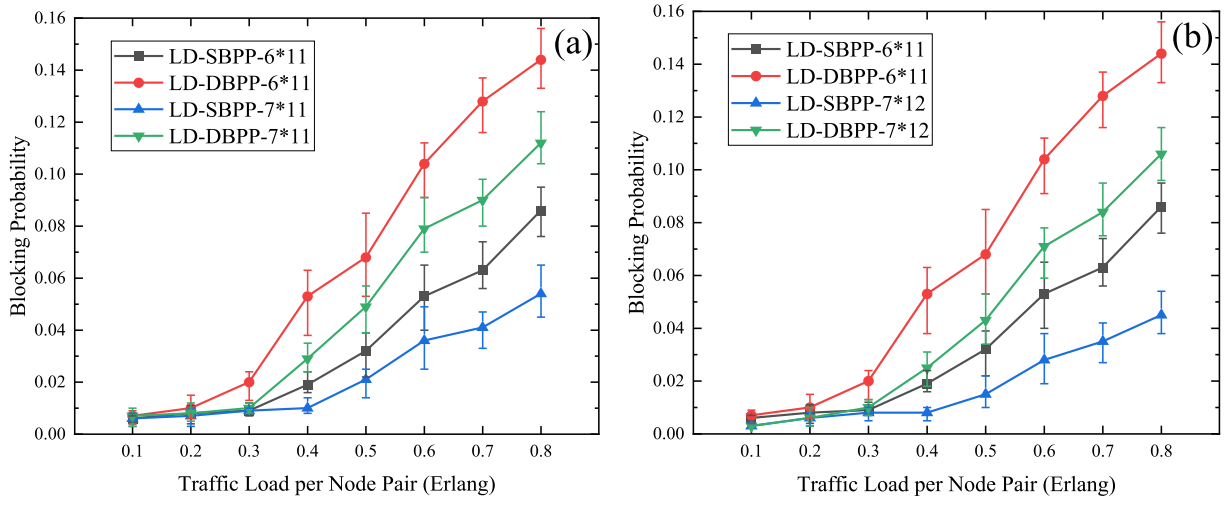


Fig. 17. Blocking probability with different satellite coverage rate: (a) the constellation with 7*11 satellites and (b) the constellation with 7*12 satellites.

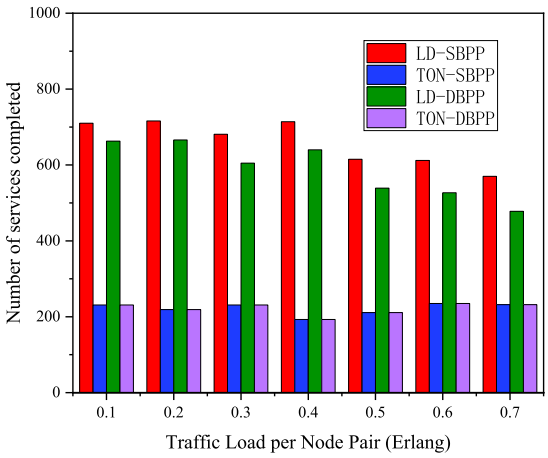


Fig. 18. Number of services completed by TON vs. number of services completed by SON.

the LD-DBPP scheme, the average number of link-switching for the $S = 5$ condition is 24.4 % higher than that for $S = 3$, and 111.9 % higher than that for $S = 1$.

5.6. Blocking Probability with Added Satellite Coverage Rate

To evaluate the impact of satellite coverage rate, we add some satellites. Satellite coverage rate is defined as the average value of the number of visible satellites at a GS. We improve satellite coverage rate by adding one orbit. The original constellation has 6*11 satellites, and now it is increased to 7*11. With the increase in the number of satellites, the average satellite coverage rate of all GSs increases from 1.46 to 1.73. The blocking probability result is shown in Fig. 17(a). The addition of satellites reduces the average blocking probability by 33.3 % and 28.1 % for LD-SBPP and LD-DBPP schemes, respectively.

On this basis, one additional satellite is added to each orbit, making the satellite coverage rate to be 1.90. The blocking probability result is shown in Fig. 17(b). Compared with the original constellation, the average blocking probability is reduced by 46.4 % and 34.8 % for LD-SBPP and LD-DBPP schemes, respectively. In essence, increasing the satellite coverage is equivalent to increasing the satellite-to-ground capacity. It can be concluded that the higher the satellite coverage rate, the lower the blocking probability.

5.7. Cost and service completion rates

From a cost perspective, SON is expensive. However, the survivability of a TON in the face of all types of major catastrophic events relies heavily on its connectivity. Network islanding may occur in extreme cases, such as major natural disasters, resulting in a complete loss of TON connectivity in certain areas. Therefore, we consider using SON as a backup emergency communication means to compensate for potential disruptions in such extreme scenarios. When using DBPP and SBPP, we always place TON availability as a primary consideration to minimize the additional costs SON may incur. In order to explore more clearly the potential impact of increased SON costs on communication outage rates, a detailed comparison between increased wavelength costs and reduced service blocking rates is introduced to gain insight into the strong relationship between wavelength costs and communication outage rates. As shown in Fig. 18, with the escalation of the load, both the services accomplished by TON and those achieved by incorporating SON exhibit corresponding variations. The average service duration is 200 s, the number of terrestrial network wavelengths is 80, the number of satellite network wavelengths is 12, and the number of simulated generated services is 1000. According to the simulation results, employing SON as a supplementary communication means for TON substantially enhances the service completion rate, albeit at the expense of incurring a considerable cost.

6. Conclusions

The LEO constellation can effectively complement TONs and help improve their survivability because it has the benefits of low latency and wide coverage, and it is not susceptible to terrestrial disasters. The LD-SBPP scheme in the LEO-satellite-terrestrial integrated network is proposed in this paper. The P-SCRWA algorithm is developed to handle LEO satellites' high dynamics and ensure successful handover. Numerical results show a significant increase in the survivability of the TON with the cooperation of LEO satellites. The LD-SBPP scheme has higher wavelength utilization efficiency than the LD-DBPP scheme. Adequate satellite-to-ground capacity is a prerequisite for the implementation of satellite backup protection mechanisms. The larger the maximum number of link-switching allowed per request, the higher the complexity of the algorithm. In the case of limiting the maximum number of link-switching allowed, link duration should be given priority.

Author contributions

Xin Li and Wei Xu were jointly responsible for the proposed scheme and algorithms. Wei Xu was responsible for the completion of the first draft of the manuscript. Yu Liu, Chenyu Zhao, and Shanguo Huang were responsible for the review, modification and editing of the manuscript. All authors read and contributed to the manuscript.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

This work was supported in part by the National Natural Science Foundation of China (Nos. 62171050 and 62125103) and the Fund of State Key Laboratory of Information Photonics and Optical Communications (Beijing University of Posts and Telecommunications) (No. IPOC2021ZT15), P. R. China.

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